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Inventor(s)

BAERD; Henri et al.

POWER SUPPLY DEVICE AND ASSOCIATED VARIABLE FREQUENCY DRIVE AND METHOD

Abstract

Provided is a power supply device for a variable frequency drive, a drive system and a method for supplying the same, the power supply device includes a switch and an energy storage device having an additional impedance and an energy storage means, a loading means to load the energy storage means when the switch is open so that the voltage of the energy storage means is equal to a loading voltage value equal to the sum of a DC maximal value and a voltage threshold, and maintain the load of the energy storage means when the switch is open so that the voltage of the energy storage means is equal to the loading voltage value, and a control means to close the switch when a DC link voltage of the variable frequency drive is equal or smaller than a DC minimal value minus a predetermined voltage value.

Inventors: BAERD; Henri (Villebon-sur-Yvette, FR), ELENJICKAIL ABDUL AZEEZ; Thufail (Villebon-sur-Yvette, FR), GRUAU; Nicolas (Villebon-sur-Yvette, FR)

Applicant: GE ENERGY POWER CONVERSION TECHNOLOGY LIMITED
(WARWICKSHIRE, GB)

Family ID: 89984601

Assignee: GE ENERGY POWER CONVERSION TECHNOLOGY LIMITED
(WARWICKSHIRE, GB)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to European Patent Application Serial Number EP24158313.7, filed Feb. 19, 2024, which is herein incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to the field of power supplies.

[0003] The present invention relates more particularly to a power supply device for a variable frequency drive, a variable frequency drive comprising such a power supply device and a method for operation such a power supply device.

BACKGROUND

[0004] Generally, a variable frequency drive VFD commands an electric machine to generate a torque from a grid delivering electric power supply.

[0005] The VFD comprises for example a rectifier connected to the grid, a direct current DC link supplied by the rectifier, and a power inverter supplied by the DC link. The power inverter supplies the electric machine with power.

[0006] The electric machine works in motor mode to drive, for example a turbomachine which may be a compressor or a pump.

[0007] The driven turbomachine may be used to implement a critical process requiring that the electric machine permanently delivers a torque having a minimal no nil value to avoid a risk of trip of the driven turbomachine.

[0008] Grid disturbances may cause short voltage dips, for example between 20 ms and 200 ms, so that the voltage of the DC link decreases.

[0009] The VFD may deliver not enough power to the electric machine to drive the turbomachine during voltage dips and may loss controllability of the electric machine.

[0010] The torque delivered by the electric machine during voltage dips decreases which may cause a trip of the driven turbomachine.

[0011] To continue to supply the electric machine to generate enough torque during voltage dips, it is known to supply the DC link with energy stored in energy storage means.

[0012] The energy storage means are connected to an auxiliary power converter.

[0013] The auxiliary power converter is connected to the DC link and supplies the DC link during voltage dips.

[0014] However, the implementation of the auxiliary power converter increases the size, the weight, and the cost of the VFD.

[0015] Further, the implementation of the auxiliary power converter deteriorates the reliability of the VFD.

[0016] It is also known to directly connect the energy storage means to the DC link, without an auxiliary power converter.

[0017] The storage capacity of the energy storage means may be determined so that the voltage of the DC link remains noticeably constant.

[0018] As the power delivered by the power inverter is for example equal to 10 MW, the energy storage means need to be oversized to store a large amount of energy increasing the size, the weight, and the cost of the VFD.

[0019] When the energy storage means are directly connected to the DC link and in order not to oversize the energy storage means, it is also known to oversize the VFD and the electric machine so that the electric machine delivers the minimal required torque to avoid tripping of the turbomachine when the DC link voltage reaches a minimal value at the end of a predetermined duration of a voltage dip.

[0020] However, oversizing the VFD and the electric machine increases the size, the weight, and the cost of the VFD and the electric machine.

[0021] It is therefore proposed to provide a method and power supply device to overcome wholly or partially this disadvantage.

SUMMARY

[0022] In view of the foregoing, provided is a method for supplying a variable frequency drive including a DC link including a first bus and a second bus, and a power inverter connected to the DC link, a power supply device including a switch and an energy storage device, the energy storage device including an additional impedance and energy storage means, a first connection of the energy storage device being connected to the first bus and a second connection of the energy storage device being connected to a first end of the switch, the second end of the switch being connected to the second bus, the DC link supplying the power inverter with a DC voltage equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value, the switch being open so that no current flows through the switch. The method includes loading the energy storage means when the switch is open so that the voltage of the energy storage means is equal to a loading voltage value equal to the sum of the DC maximal value and a voltage threshold, maintaining the load of the energy storage means when the switch is open so that the voltage of the energy storage means is equal to the loading voltage value, and closing the switch when the DC link voltage is equal or smaller than the DC minimal value minus a predetermined voltage value to supply the DC link with an energy flow from the energy storage device, the energy flow value being positive as long as the energy storage device supplies the DC link.

[0023] According to an embodiment, the method further includes opening the switch when the energy flow supplied by the energy storage device is smaller or equal to a positive predetermined value.

[0024] Advantageously, the method includes opening the switch when the energy flow supplied by the energy storage device is equal or smaller than a negative predetermined value.

[0025] Another embodiment of the invention relates to a power supply device for variable frequency drive including a power inverter connected to a DC link, the DC link including a first bus and a second bus and being configured to supply the power inverter with a DC voltage equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value, the power supply device comprising a switch and an energy storage device, the energy storage device including an additional impedance and energy storage means, and a first connection of the energy storage device being configured to be connected to the first bus and a second connection of the energy storage device being connected to a first end of the switch, the second end of the switch being configured to be connected to the second bus.

[0026] The power supply device includes a loading means configured to load the energy storage means when the switch is open so that the voltage of the energy storage means is equal to a loading voltage value equal to the sum of the DC maximal value and a voltage threshold, and maintain the load of the energy storage means when the switch is open so that the voltage of the energy storage means is equal to the loading voltage value, and control means configured to close the switch when

the DC link voltage is equal or smaller than the DC minimal value minus a predetermined voltage value to supply the DC link with an energy flow from the energy storage device, the energy flow value being positive as long as the energy storage device supplies the DC link.

[0027] Advantageously, the switch is configured to be opened when the energy flow supplied by the energy storage device is smaller or equal to a positive predetermined value.

[0028] According to an embodiment, the switch is configured to be opened when the energy flow supplied by the energy storage device is equal or smaller than a negative predetermined value.

[0029] According to an embodiment, the switch includes a static bidirectional switch.

[0030] According to an embodiment, the switch includes a static unidirectional switch.

[0031] Advantageously, the switch includes at least one thyristor, the anode of the thyristor being connected to the first end of the switch and the cathode of the thyristor being connected to the second end of the switch.

[0032] Preferably, the energy storage device includes a supercapacitor.

[0033] Advantageously, the energy storage means includes at least one capacitor connected in series with the additional impedance.

[0034] Preferably, the energy storage means includes at least one battery connected in series with the additional impedance.

[0035] Another embodiment of the invention relates to a variable frequency drive including a power inverter connected to a DC link, the DC link including a first bus and a second bus and being configured to supply the power inverter with a DC voltage equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value, and a power supply device as defined above, connected to the first bus and a second bus.

[0036] Advantageously, the variable frequency drive further comprises a rectifier configured to be connected to a grid and connected to the DC link to supply the DC link with the DC voltage.

[0037] According to an embodiment, the rectifier is a diode rectifier.

[0038] Another embodiment of the invention relates to a drive system including a variable frequency drive as defined above, and an electric machine connected to the power inverter of the variable frequency drive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] Other characteristics and advantages of the invention will emerge on reading the following description of embodiments of the invention, provided solely by way of non-limiting examples and with reference to the drawings in which:

[0040] FIG. 1 illustrates schematically an embodiment of a drive system according to the invention,

[0041] FIG. 2 illustrates schematically a first embodiment of a method to control the drive system according to the invention,

[0042] FIG. 3 illustrates schematically a second embodiment of the method to control the drive system according to the invention,

[0043] FIG. 4 illustrates schematically a first embodiment of a switch according to the invention,

[0044] FIG. 5 illustrates schematically a second embodiment of the switch according to the invention,

[0045] FIG. 6 illustrates schematically a third embodiment of the method to control the drive system according to the invention,

[0046] FIG. 7 illustrates schematically a third embodiment of the switch according to the invention,

[0047] FIG. 8 illustrates schematically a first embodiment of a bridge for a rectifier according to

the invention,

[0048] FIG. **9** illustrates schematically a second embodiment of the bridge according to the invention, and

[0049] FIG. **10** illustrates an embodiment of an energy storage device according to the invention.

DETAILED DESCRIPTION

[0050] FIG. **1** illustrates schematically an embodiment of a drive system **1**.

[0051] The drive system **1** comprises a variable frequency drive VFD **2** and an electric machine **3** connected to the VFD **2**.

[0052] The electric machine **3** comprises a rotor shaft **4** connected to a load **5**.

[0053] The load **5** comprises for embodiment a turbomachine, for example a pump or a compressor.

[0054] The VFD **2** is connected to a grid G through a transformer **6**.

[0055] A primary circuit **6a** of the transformer **6** is connected to the grid G and a secondary circuit **6b** of the transformer **6** is connected to the VFD.

[0056] The grid G may comprise N phase(s) and the electric machine **3** may comprise M phases, N and M being non nil integers, M being at least equal to two.

[0057] It is assumed in the following that N and M are equal to three, the grid G being a triphasic grid and the electric machine **3** being a triphasic machine, and that each circuit of the primary circuit **6a** and secondary circuit **6b** comprises three phases.

[0058] Each circuit of the primary circuit **6a** and secondary circuit **6b** may comprise more or less than three phases.

[0059] The VFD **2** comprises a rectifier **7**, a direct current DC link **8** and a power inverter **9**.

[0060] The DC link **8** comprises a first bus **8a**, a second bus **8b** and a capacitor **8c** connected to the first and second buses **8a**, **8b**.

[0061] As the secondary circuit **6b** comprises three phases, the rectifier **7** comprises three inputs **10**, **11**, **12**, each input being connected to a phase of the secondary circuit **6b**.

[0062] In another embodiment, the grid G is connected directly to the inputs **10**, **11**, **12** of the rectifier **7**, without the transformer **6**.

[0063] The rectifier **7** further comprises a first output **13** connected to a first end of the second bus **8b** and a second output **14** connected to a first end of the first bus **8a**.

[0064] The rectifier **7** may comprise a same number of bridges **15**, **16**, **17** as the number of inputs **10**, **11**, **12** of the rectifier **7**.

[0065] Each bridge **15**, **16**, **17** comprises a first connection **15a**, **16a**, **17a** connected to an input **10**, **11**, **12** of the rectifier **7**, a second connection **15b**, **16b**, **17b** connected to the first output **13** of the rectifier **7** and a third connection **15c**, **16c**, **17c** connected to the second output **14** of the rectifier **7**.

[0066] When the secondary circuit **6b** of the transformer **6** comprises more than three phases, the person skilled in the art may arrange the structure of the rectifier **7** with series or parallel connections of bridges as known from the prior art.

[0067] A second end of the first and second buses **8a**, **8b** are connected to inputs **19**, **18** of the power inverter **9**.

[0068] The power inverter **9** comprises a same number of outputs **20**, **21**, **22** as the number of phases M of the electric machine **3**, each output **20**, **21**, **22** being connected to a phase of the machine **3**.

[0069] The drive system **1** further comprises a power supply device **23** comprising a switch **24**, an energy storage device **25** including energy storage means **26** and an additional impedance **27**, loading means **28** and control means **29**.

[0070] The energy storage device **25** comprises for example a supercapacitor having an embedded impedance, the embedded impedance being the additional impedance **27**.

[0071] The energy storage device **25** comprises a first connection **25a** connected to the first bus **8a**, a second connection **25b** connected to a first end **24a** of the switch **24**, a third connection **25c** and a

fourth connection **25d** both connected to the loading means **28**.

[0072] The second end **24b** of the switch **24** is connected to the second bus **8b**.

[0073] The loading means **28** comprise for example an auxiliary power converter to deliver a DC current to load the energy storage means **26** when the switch **24** is open and to maintain the load of the energy storage means when the switch **24** is open so that the voltage of the energy storage means **26** is equal to a loading voltage value.

[0074] The control means **29** are intended to command the switch **24** and may be intended to command the loading means **28**.

[0075] The control means **29** comprise for example a processing unit.

[0076] FIG. **2** illustrates schematically a first example of a method to control the drive system **1**.

[0077] It is assumed that the grid **G** supplies the rectifier **7** through the transformer **6** with a triphasic alternating current AC and that the switch **24** is open so that no current flows through the switch **24**.

[0078] The rectifier **7** rectifies the AC current and supplies the DC link **8** so that the DC voltage between the first and second bus **8a**, **8b** is equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value.

[0079] The DC minimal value is equal to the minimal voltage value delivered by the grid **G** in absence of a voltage dip multiplied by the ratio of voltage transformation of the transformer **6** and the rectification factor of the rectifier **7**.

[0080] The DC maximal value is equal to the maximal voltage value delivered by the grid **G** multiplied by the ratio of voltage transformation of the transformer **6** and the rectification factor of the rectifier **7**.

[0081] During a step **30**, the switch **24** is open and the loading means **28** load the energy storage means **26** so that the voltage of the energy storage means **26** is equal to the loading voltage value and maintain the load of the energy storage means **26** so that the voltage of the energy storage means **26** is equal to the loading voltage value.

[0082] The loading voltage value is equal to the sum of the DC maximal value and a voltage threshold.

[0083] The voltage threshold is for example equal to 10% of the DC maximal value.

[0084] As long as the voltage of the DC link **8** is bigger than the DC minimal value minus a predetermined voltage value (step **31**), the method goes back to step **30**.

[0085] The predetermined voltage value may be nil.

[0086] When the voltage of the DC link **8** is equal or smaller than the DC minimal value minus the predetermined voltage value (step **31**), during a step **32**, the control means **29** close the switch **24** so that the energy storage device **25** supplies the DC link **8**.

[0087] The voltage of the DC link **8** equal or smaller than the DC minimal value minus the predetermined voltage value is representative of a voltage dip of the grid **G**.

[0088] The energy storage device **25** supplies the DC link **8** with an energy flow.

[0089] The control means **29** monitor the energy flow.

[0090] The energy flow value is positive as long as the energy storage device **25** supplies the DC link **8**.

[0091] When the energy flow value is positive or nil, the energy storage device **25** operates as a source of energy and supplies the DC link **8**.

[0092] When the energy flow value is negative, the energy storage device **25** operates as a receiver of energy and is supplied by the DC link **8**.

[0093] The DC link **8** is supplied by the energy storage device **25** having a voltage at least equal to the DC minimal voltage.

[0094] As the energy storage device **25** comprises the additional impedance **27**, the inrush current generated during the closing of the switch **24** by the connection of the energy storage means **26**

having a voltage equal to the loading voltage and the DC link **8** having a voltage equal or smaller than the DC minimal value minus the predetermined voltage value is limited.

[0095] The energy storage device **25** may be designed so that the additional impedance **27** is chosen so that the inrush current is for example equal to maximal two times the nominal current.

[0096] As the inrush current is for example equal to maximal two times the nominal current, the power inverter **9**, the capacitor **8**, the energy storage device **25** and the switch **24** are made from standard components reducing the complexity to realize the power inverter **9**, the capacitor **8**, the energy storage device **25** and the switch **24** and reducing the costs.

[0097] As a supercapacitor comprises intrinsically an embedded impedance, the arrangement of power modules forming the supercapacitor may be determined so that the additional impedance **27** of the energy storage device **25** which is the embedded impedance of the supercapacitor is designed to limit the inrush current to the chosen inrush current value.

[0098] The energy storage means **26** are further designed so that the energy stored is enough to supply the power inverter **9** during the duration of at least one voltage dip, for example for 200 ms.

[0099] The amount of energy stored in the energy storage means **26** is determined from the duration of a reference voltage dip, the number of reference voltage dip(s) before charging the energy storage means **26** and the power supplied to the electric machine **3** so that the torque generated by the electric machine **3** is enough to drive the load **5** without dysfunction.

[0100] As long as the energy storage device **25** supplies the DC link **8** with the energy flow having a positive energy flow value larger than a positive predetermined value (step **33**), the method goes back to step **32**.

[0101] The positive predetermined value is for example equal to 5% of the nominal power of the electric machine **3**.

[0102] The energy flow is for example determined by the control means **29** from measurements of the current flowing in the switch **24** and the voltage on the DC link **8** or the voltage on the energy storage device **25**.

[0103] The current flowing in the switch **24** may be measured by a current sensor connected to the control means **29**. The voltage on the DC link **8** or the voltage on the energy storage device **25** may be measured by a voltage sensor connected to the control means **29**.

[0104] In another embodiment, the energy flow may be determined from a model of the energy storage device **25**, the evolution of a voltage and the instantaneous value of the said voltage, the said voltage being the voltage on the DC link **8** or the voltage on the energy storage device **25**.

[0105] In another embodiment, the energy flow may be determined from a model of the energy storage device **25**, the evolution of the current flowing in the switch **24** and the instantaneous value of the current flowing in the switch **24**.

[0106] When the energy flow supplied by the energy storage device **25** is smaller or equal to the positive predetermined value (step **33**), in a step **34**, the switch **24** is opened and the method goes back to step **30**.

[0107] The loading means **28** load the energy storage means **26** when the switch **24** is open and maintain the load of the energy storage means when the switch **24** is open so that the voltage of the energy storage means **26** is equal to the loading voltage value.

[0108] The electric machine **3** is supplied by the grid G.

[0109] During a voltage dip, the electric machine **3** is supplied by the energy storage device **25** having a voltage at least equal to the DC minimal voltage so that the DC link voltage is between at least the DC minimal voltage and the DC maximal voltage so that the power inverter **9** and the electric machine **3** need not to be oversized to run at a DC link voltage smaller than the DC minimal voltage.

[0110] The auxiliary power converter **28** is designed to load the energy storage means **26** requiring less power than to supply the DC link **8** so that the size, the weight and the cost of the auxiliary power converter **28** is reduced compared to the auxiliary power converter known from the prior art.

[0111] During steps **32** and **33**, the loading means **28** may remain loading the energy storage means **26** or may stop to load the energy storage means **26**.

[0112] FIG. **3** illustrates schematically a second embodiment of the method to control the drive system **1**.

[0113] It is assumed that the grid **G** supplies the rectifier **7** through the transformer **6** with a triphasic alternating current AC and that the switch **24** is open so that no current flows through the switch **24**.

[0114] The rectifier **7** rectifies the AC current and supplies the DC link **8** so that the DC voltage between the first and second bus **8a**, **8b** is equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value.

[0115] The DC minimal value is equal to the minimal voltage value delivered by the grid **G** in absence of a voltage dip multiplied by the ratio of voltage transformation of the transformer **6** and the rectification factor of the rectifier **7**.

[0116] The DC maximal value is equal to the maximal voltage value delivered by the grid **G** multiplied by the ratio of voltage transformation of the transformer **6** and the rectification factor of the rectifier **7**.

[0117] The control means **29** monitor the energy flow.

[0118] The second embodiment of the method comprises the steps **30**, **31** and **32** as described above.

[0119] The second embodiment of the method differs from the first embodiment of the method illustrated on FIG. **2** in that the step **33** is replaced by a step **35** and that the step **34** is replaced by a step **36**.

[0120] In step **35**, the energy flow delivered by the energy storage device **25** is compared to a negative predetermined value.

[0121] The absolute value of the negative predetermined value is for example equal to **20%** of the nominal power of the rectifier **7**.

[0122] During steps **32** and **35**, the loading means **28** may remain loading the energy storage means **26** or may stop to load the energy storage means **26**.

[0123] As long as the energy flow delivered by the energy storage device **25** is bigger than the negative predetermined value, the method goes back to step **32**. When the energy flow delivered by the energy storage device **25** has a negative value, the grid **G** supplies the energy storage device **25**.

[0124] When the energy flow delivered by the energy storage device **25** is equal or smaller than the negative predetermined value, the method continues at step **36**.

[0125] During the step **36**, the control means **29** open the switch **24**.

[0126] The loading means **28** load the energy storage means **26** when the switch **24** is open and maintain the load of the energy storage means when the switch **24** is open so that the voltage of the energy storage means **26** is equal to the loading voltage value.

[0127] The switch **24** is opened when the energy flow delivered by the energy storage device **25** is equal or smaller than the negative predetermined value so that the energy storage means **26** are partially loaded by the rectifier **7** supplying more power than the auxiliary power converter **28** to load faster the energy storage means **26** during a duration between a first instant when the energy flow delivered by the energy storage device **25** is nil and a second instant when the energy flow delivered by the energy storage device **25** is equal or smaller than the negative predetermined value.

[0128] FIG. **4** illustrates schematically a first embodiment of the switch **24**.

[0129] The switch **24** comprises a static unidirectional switch **400**.

[0130] The static unidirectional switch **400** may comprise at least one transistor **40**.

[0131] The switch **24** may comprise a plurality of transistors connected in series.

[0132] The transistor **40** may be a transistor, for example an insulated-gate bipolar transistor IGBT

with reverse blocking capability.

[0133] The collector of the transistor **40** is connected to the first end **24a** of the switch **24** and the emitter of the transistor **40** is connected to the second end **24b** of the switch **24**.

[0134] The gate **41** is connected to the control means **29**.

[0135] The transistor **40** may be implemented in the first embodiment of the method illustrated in FIG. 2.

[0136] During the step **32**, the control means **29** command the gate of the transistor **40** to close the transistor **40** so that the energy storage device **25** supplies the DC link **8**.

[0137] In the first embodiment of the method, when the energy flow supplied by the energy storage device **25** is smaller or equal to the positive predetermined value, in the step **34**, the transistor **40** is opened so that no current flows through the transistor **40** following the reception of a command signal of the control means **29** on the gate of the transistor **40**.

[0138] The transistor **40** is not implemented in the second embodiment of the method to control the drive system **1** illustrated in FIG. 3.

[0139] FIG. 5 illustrates schematically a second embodiment of the switch **24**.

[0140] The switch **24** comprises a static bidirectional switch **500** including a first transistor **50**, a second transistor **51**, a first diode **52** and a second diode **53**.

[0141] Each transistor of the first and second transistors **50**, **51** may be an insulated-gate bipolar transistor IGBT.

[0142] The collector of the first transistor **50** is connected to the first end **24a** of the switch **24** and the emitter of the first transistor **50** is connected to the emitter of the second transistor **51**.

[0143] The collector of the second transistor **51** is connected to the second end **24b** of the switch **24**.

[0144] The gates **54**, **55** of the first and second transistors **50**, **51** are connected to the control means **29**.

[0145] The cathode of the first diode **52** is connected to the first end **24a** of the switch **24**, the anode of the first diode **52** is connected to the anode of the second diode **53** and the emitter of the first transistor **50**.

[0146] The cathode of the second diode **53** is connected to the collector of the second transistor **51** and to the second end **24b** of the switch **24**.

[0147] The static bidirectional switch **500** may be implemented in the first embodiment of the method illustrated in FIG. 2 and in the second embodiment of the method illustrated in FIG. 3.

[0148] In steps **30** and **31**, the control means **29** command the gates **54**, **55** of the first and second transistors **50**, **51** to open the first and second transistors **50**, **51** so that no current flows through the transistors **50**, **51**.

[0149] In the first embodiment of the method, during the step **32**, the control means **29** command the gate **54** of the first transistor **50** to close the first transistor **50** so that the energy storage device **25** supplies the DC link **8**, and the gate **55** of the second transistor **51** so that the second transistor **51** remains open.

[0150] When the energy flow supplied by the energy storage device **25** is smaller or equal to the positive predetermined value (step **33**), in the step **34**, the control means **29** command the gate **54** of the first transistor **50** to open the first transistor **50**, and the gate **55** of the second transistor **51** so that the second transistor **51** remains open.

[0151] In the second embodiment of the method, during the step **32**, the control means **29** command the gate **54** of the first transistor **50** to close the first transistor **50** and the gate **55** of the second transistor **51** to close the second transistor **51** so that the energy storage device **25** supplies the DC link **8**.

[0152] When the energy flow supplied by the energy storage device **25** is equal or smaller than the negative predetermined value (step **35**), in the step **36**, the control means **29** command the gate **54** of the first transistor **50** to open the first transistor **50** and the gate **55** of the second transistor **51** to

open the second transistor **51**.

[0153] FIG. **6** illustrates a third embodiment of the method to control the drive system **1**.

[0154] It is assumed that the grid **G** supplies the rectifier **7** through the transformer **6** with a triphasic alternating current AC and that the switch **24** is open so that no current flows through the switch **24**.

[0155] The rectifier **7** rectifies the AC current and supplies the DC link **8** so that the DC voltage between the first and second bus **8a**, **8b** is equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value.

[0156] The DC minimal value is equal to the minimal voltage value delivered by the grid **G** in absence of a voltage dip multiplied by the ratio of voltage transformation of the transformer **6** and the rectification factor of the rectifier **7**.

[0157] The DC maximal value is equal to the maximal voltage value delivered by the grid **G** multiplied by the ratio of voltage transformation of the transformer **6** and the rectification factor of the rectifier **7**.

[0158] The control means **29** monitor the energy flow.

[0159] In this embodiment of the method, the switch **24** comprises at least one thyristor **70** illustrated in FIG. **7**.

[0160] The switch **24** may comprise a plurality of thyristors connected in series.

[0161] The anode of the thyristor **70** is connected to the first end **24a** of the switch **24** and the cathode of the thyristor **70** is connected to the second end **24b** of the switch **24**.

[0162] The gate of the thyristor **70** is connected to the control means **29**.

[0163] The third embodiment of the method comprises the steps **30**, **31**, **32** as described above.

[0164] During the step **32**, the control means **29** command the gate of the thyristor **70** to close the thyristor **70** so that the energy storage device **25** supplies the DC link **8**.

[0165] As long as a current having a positive value greater than the thyristor holding current flows from the second connection **25b** of the energy storage device **25** to the second bus **8b** of the DC link **8** (step **60**), the thyristor **70** remains intrinsically closed.

[0166] When the current flowing from the second connection **25b** of the energy storage device **25** to the second bus **8b** of the DC link **8** (step **60**) is smaller than the thyristor holding current, in a step **61**, the thyristor **70** opens so that no current flows through the thyristor **70** and the method goes back to step **30**.

[0167] The loading means **28** load the energy storage means **26** when the switch **24** is open and maintain the load of the energy storage means when the switch **24** is open so that the voltage of the energy storage means **26** is equal to the loading voltage value.

[0168] The structure of the thyristor **70** allows to command the opening of the thyristor **70** automatically without a command of the control means **29**.

[0169] FIG. **8** illustrates schematically a first embodiment of the bridge **15**, **16**, **17** of the rectifier **7**.

[0170] The rectifier **7** is a diode rectifier.

[0171] The bridge **15**, **16**, **17** comprises a first diode **80** and a second diode **81** connected in series.

[0172] The cathode of the first diode **80** is connected to the second connection **15b**, **16b**, **17b** of the bridge **15**, **16**, **17**, and the anode of the first diode **80** is connected to the first connection **15a**, **16a**, **17a** of the bridge **15**, **16**, **17**.

[0173] The cathode of the second diode **81** is connected to the first connection **15a**, **16a**, **17a** of the bridge **15**, **16**, **17**, and the anode of the second diode **81** is connected to the third connection **15c**, **16c**, **17c** of the bridge **15**, **16**, **17**.

[0174] The bridge **15**, **16**, **17** comprising the diodes **80**, **81** is not commanded.

[0175] FIG. **9** illustrates schematically a second embodiment of the bridge **15**, **16**, **17** of the rectifier **7**.

[0176] The rectifier **7** may be a controlled rectifier.

[0177] The bridge **15, 16, 17** comprises a first transistor **90** and a second transistor **91** connected in series.

[0178] The bridge **15, 16, 17** further comprises a first diode **92** and a second diode **93**.

[0179] Each transistor of the first and second transistors **90, 91** may be an insulated-gate bipolar transistor IGBT.

[0180] The collector of the first transistor **90** is connected to the second connection **15b, 16b, 17b** of the bridge **15, 16, 17**, the emitter of the first transistor **90** is connected to the first connection **15a, 16a, 17a** of the bridge **15, 16, 17**.

[0181] The collector of the second transistor **91** is connected to the first connection **15a, 16a, 17a** of the bridge **15, 16, 17**, and the emitter of the second transistor **91** is connected to the third connection **15c, 16c, 17c** of the bridge **15, 16, 17**.

[0182] The anode of the first diode **92** is connected to the first connection **15a, 16a, 17a** of the bridge **15, 16, 17** and the cathode of the first diode **92** is connected to the second connection **15b, 16b, 17b** of the bridge **15, 16, 17**.

[0183] The anode of the second diode **93** is connected to the to the third connection **15c, 16c, 17c** of the bridge **15, 16, 17** and the cathode of the second diode **93** is connected to the first connection **15a, 16a, 17a** of the bridge **15, 16, 17**.

[0184] The gate of the first and second transistors **90, 91** is connected to a control unit of the VFD **2** (not represented).

[0185] FIG. **10** illustrates schematically another embodiment of the energy storage device **25**.

[0186] The energy storage means **26** comprise at least one capacitor **100** or at least one battery **101**.

[0187] As the embedded impedance of the capacitor **100** or the battery **101** is too low to limit the inrush current to the chosen inrush current value, an impedance **102** is connected in series with the capacitor **100** or the battery **101**, the impedance **102** being the additional impedance **27**.

[0188] A first end of the impedance **102** is connected to the third connection **25c** of the energy storage device **25** and the second end of the impedance **102** is connected to the second connection **25b** of the energy storage device **25**.

[0189] In another embodiment, the first end of the impedance **102** is connected to the fourth connection **25d** of the energy storage device **25** and the second end of the impedance **102** is connected to the first connection **25a** of the energy storage device **25**.

Claims

1. A method for supplying a variable frequency drive comprising a DC link including a first bus and a second bus, and a power inverter connected to the DC link, a power supply device comprising a switch and an energy storage device, the energy storage device including an additional impedance and an energy storage means, a first connection of the energy storage device being connected to the first bus and a second connection of the energy storage device being connected to a first end of the switch, the second end of the switch being connected to the second bus, the DC link supplying the power inverter with a DC voltage equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value, the switch being open so that no current flows through the switch, the method comprises: loading the energy storage means when the switch is open so that the voltage of the energy storage means is equal to a loading voltage value equal to the sum of the DC maximal value and a voltage threshold, maintaining the load of the energy storage means when the switch is open so that the voltage of the energy storage means is equal to the loading voltage value, and closing the switch when the DC link voltage is equal or smaller than the DC minimal value minus a predetermined voltage value to supply the DC link with an energy flow from the energy storage device, the energy flow value being positive as long as the energy storage device supplies the DC link.

2. The method according to claim 1, comprising opening the switch when the energy flow supplied by the energy storage device is smaller or equal to a positive predetermined value.
 3. The method according to claim 1, comprising opening the switch when the energy flow supplied by the energy storage device is equal or smaller than a negative predetermined value.
 4. A power supply device for variable frequency drive comprising a power inverter connected to a DC link, the DC link including a first bus and a second bus and being configured to supply the power inverter with a DC voltage equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value, the power supply device comprising: a switch and an energy storage device, the energy storage device including an additional impedance, and an energy storage means, and a first connection of the energy storage device being configured to be connected to the first bus and a second connection of the energy storage device being connected to a first end of the switch, the second end of the switch being configured to be connected to the second bus, the power supply device further comprises: a loading means configured to load the energy storage means when the switch is open so that the voltage of the energy storage means is equal to a loading voltage value equal to the sum of the DC maximal value and a voltage threshold, and maintain the load of the energy storage means when the switch is open so that the voltage of the energy storage means is equal to the loading voltage value, and a control means configured to close the switch when the DC link voltage is equal or smaller than the DC minimal value minus a predetermined voltage value to supply the DC link with an energy flow from the energy storage device, the energy flow value being positive as long as the energy storage device supplies the DC link.
 5. The power supply device according to claim 4, wherein the switch is configured to be opened when the energy flow supplied by the energy storage device is smaller or equal to a positive predetermined value.
 6. The power supply device according to claim 5, wherein the switch is configured to be opened when the energy flow supplied by the energy storage device is equal or smaller than a negative predetermined value.
 7. The power supply device according to claim 6, wherein the switch comprises a static bidirectional switch.
 8. The power supply device according to claim 5, wherein the switch comprises a static unidirectional switch.
 9. The power supply device according to claim 4, wherein the switch comprises at least one thyristor, the anode of the thyristor being connected to the first end of the switch and the cathode of the thyristor being connected to the second end of the switch.
 10. The power supply device according to claim 9, wherein the energy storage device comprises a supercapacitor.
 11. The power supply device according to claim 9, wherein the energy storage means comprises at least one capacitor connected in series with the additional impedance.
 12. The power supply device according to claim 9, wherein the energy storage means comprises at least one battery connected in series with the additional impedance.
 13. A variable frequency drive comprising: a power inverter connected to a DC link, the DC link including a first bus and a second bus and being configured to supply the power inverter with a DC voltage equal to a DC nominal voltage value included between a DC minimal value and a DC maximal value, the DC maximal value being bigger than the DC minimal value, and a power supply device according to claim 12, connected to the first bus and a second bus.
 14. The variable frequency drive according to claim 13, further comprising a rectifier configured to be connected to a grid and connected to the DC link to supply the DC link with the DC voltage.
 15. A drive system comprising: a variable frequency drive according to claim 14, and an electric machine connected to the power inverter of the variable frequency drive.
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